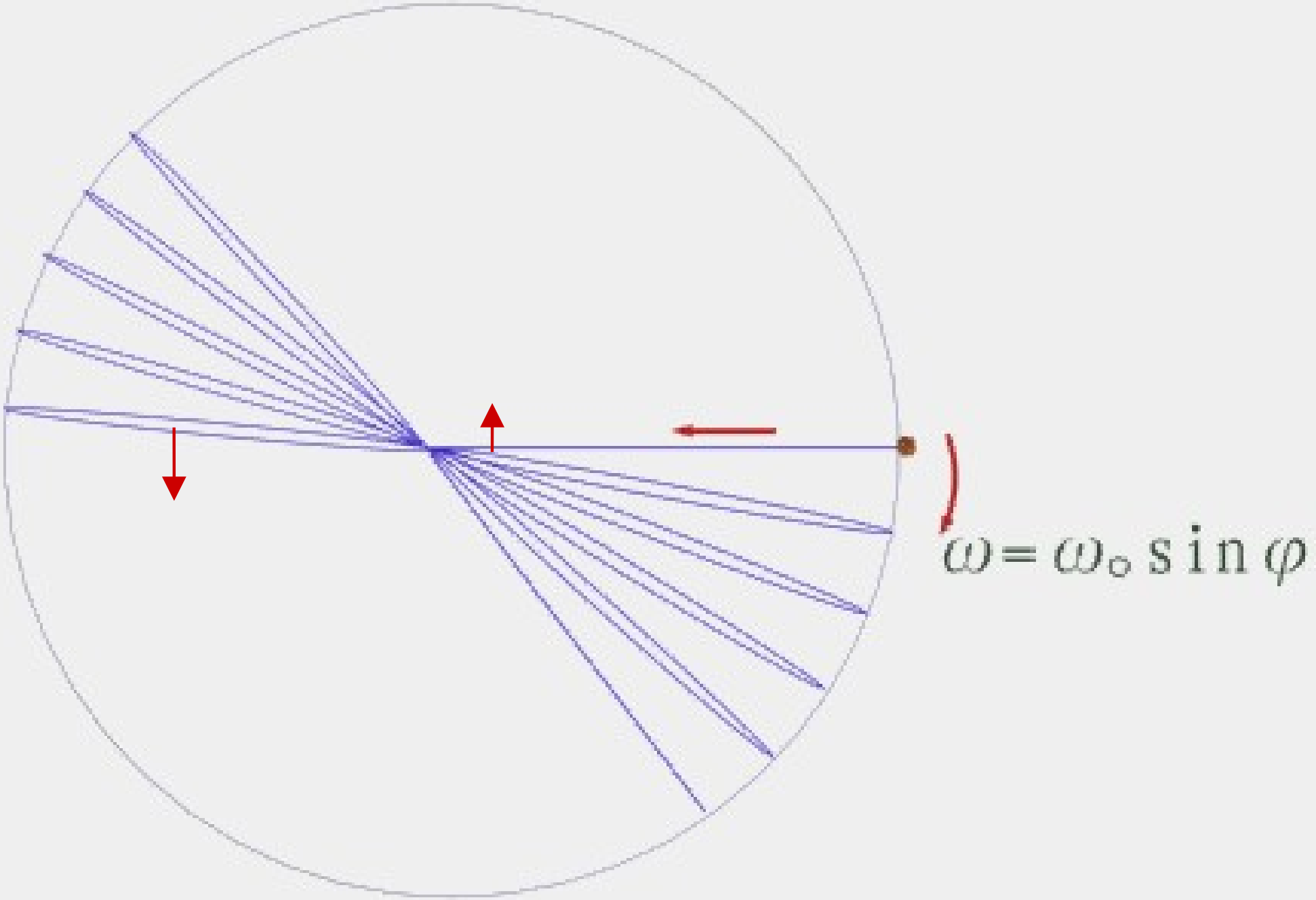

$\omega = \omega_0 \sin \varphi$